### 2.6.4 Electronic configuration of atoms

The distribution of electrons into various orbitals of an atom is called its electronic configuration. It can be written by applying the aufbau principle, Pauli exclusion principle and Hund's rule. The electronic configuration is written as $nl^x$ , where n represents the principle of quantum number, '*l*' represents the letter designation of the orbital [s(*l*=0), p (*l*=1), d(*l*=2) and f(*l*=3)] and 'x' represents the number of electron present in that orbital.

Let us consider the hydrogen atom which has only one electron and it occupies the lowest energy orbital i.e. 1s according to aufbau principle. In this case n=1; *l*=s; x=1.

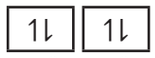
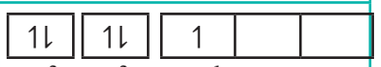
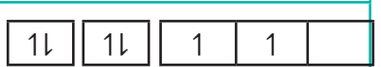
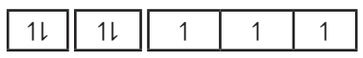
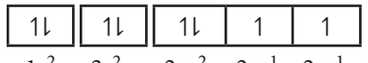
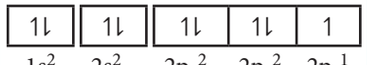
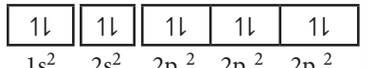
Hence the electronic configuration is $1s^1$. (read as one-ess-one).

The orbital diagram for this configuration is,




The electronic configuration and orbital diagram for the elements upto atomic number 10 are given below :

**Table 2. 4 Electronic configuration and orbital diagrams for first 10 elements**

| Element | Electronic Configuration | Orbital diagram |
|---|---|---|
| $H^1$ | $1s^1$ |  |
| $He^2$ | $1s^2$ |  |
| $Li^3$ | $1s^2\ 2s^1$ |  |
| $Be^4$ | $1s^2\ 2s^2$ |  |
| $B^5$ | $1s^2\ 2s^2\ 2p^1$ |  |
| $C^6$ | $1s^2\ 2s^2\ 2p^2$ |  |
| $N^7$ | $1s^2\ 2s^2\ 2p^3$ |  |
| $O^8$ | $1s^2\ 2s^2\ 2p^4$ |  |
| $F^9$ | $1s^2\ 2s^2\ 2p^5$ |  |
| $Ne^{10}$ | $1s^2\ 2s^2\ 2p^6$ |  |

The actual electronic configuration of some elements such as chromium and copper slightly differ from the expected electronic configuration in accordance with the Aufbau principle.



11th Std Chemistry 038-067.indd 54 06-01-2021 15:06:52